

LP3 (ISR) – Assignment 5 – F-measure and E-measure (RollNo: 43221)

Input Code:

```
import java.util.*;

public class MetricsCalculator {
    public static double calculateF1(double precision, double recall) {
        if (precision + recall == 0) {
            return 0;
        }
        return 2 * (precision * recall) / (precision + recall);
    }

    public static double calculateEMeasure(double precision, double recall,
double alpha) {
        if (precision == 0 && recall == 0) {
            return 0;
        }
        return 1 / ((alpha / precision) + ((1 - alpha) / recall));
    }

    public static void main(String[] args) {
        // Example values
        double precision = 0.75;
        double recall = 0.80;
        double alpha = 0.5;
        // Calculate F-measure
        double f1Score = calculateF1(precision, recall);
        System.out.println("F-measure (F1-score): " +
            String.format("%.2f", f1Score));
        // Calculate E-measure
        double eMeasure = calculateEMeasure(precision, recall, alpha);
        System.out.println("E-measure: " + String.format("%.2f",
            eMeasure));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Community Edition 2024.1.4\lib\idea_rt.jar=58361:C:\Program Files\JetBrains\IntelliJ IDEA
Community Edition 2024.1.4\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -
Dsun.stderr.encoding=UTF-8 -classpath
F:\BE\LP3\GroupB_Assignment_2_MetricsCalculator\out\production\GroupB_Assignment_2_Metric
sCalculator MetricsCalculator
```

F-measure (F1-score): 0.77

E-measure: 0.77

Process finished with exit code 0